

System Design Research: Rancang Bangun Web-Based COBIT 2019 Assessment System

Builds on [01-literature-review.md](#). Covers: existing-tools gap, RAG assessment, scope options, DSRM methodology, titles, RQs.

Date: 2026-09-30

Domain: SAGE (Software Engineering, Architecture, Governance)

Thesis Level: S1 Information Systems

1. Executive Conclusion

1.1 The Core Finding: A Genuine Gap in Native COBIT 2019 Assessment

The most significant finding from this research is that **ISACA publishes no Process Assessment Model (PAM) for COBIT 2019**. This is stated explicitly by ISACA itself: "There is no PAM for COBIT 2019, but Capability Maturity Model Integration (CMMI) can be used to measure capability levels" (ISACA Journal, 2021). The consequence is profound: every commercial tool that claims "COBIT 2019 support" actually provides only **content mapping** (objectives, practices, controls as reference content), not an **assessment instrument** that implements the official Fully/Largely/Partially/Not rating scale or the 1,202-activity to 231-practice to 40-objective to 5-domain aggregation chain.

This means the student's project occupies a genuinely underserved niche. The commercial market is saturated with GRC platforms that treat COBIT as one framework among dozens, but essentially empty of tools that implement the actual COBIT 2019 Performance Management (CPM) methodology. The academic literature confirms this: existing prototypes (Mandiangan 2023, Anwar and Harits 2025, Noor 2021) are isolated, undocumented in terms of scoring methodology, and none implement the full ISACA rating scale with proper aggregation.

1.2 The Rebalance: From RAG-Centric to Assessment-Centric

The previous literature review was too RAG-heavy. This iteration rebalances toward **COBIT 2019 assessment system design and development** as the primary contribution. The research hierarchy is now clear:

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COBIT 2019 assessment process and requirements
-> Assessment workflow and evidence management problems
-> Web-based assessment support system (rancang bangun)
-> RAG/AI as ONE supporting feature (e.g., recommendations)
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The student's thesis is about **building a system that implements the COBIT 2019 assessment methodology correctly**, not about advancing RAG techniques. RAG may assist with evidence retrieval or recommendation drafting, but the core contribution is the assessment system itself.

1.3 What the Prototype Already Does Well

The existing prototype (8 pages: Dashboard, AssessmentSetup, AssessmentWorkspace, CapabilityResult,

GapAnalysis, EvidenceFindings, Recommendations, Landing) already covers the core assessment workflow. The type system (`src/types/index.ts`) correctly models COBIT domains, capability levels (0-5), response ratings (N/P/L/F), evidence, assessments, and gap analysis. This is a solid foundation that maps directly to the COBIT 2019 assessment process.

1.4 The Honest Assessment of RAG's Role

After evaluating each potential RAG use case against the criteria of necessity, feasibility, and risk, the conclusion is that **RAG is not necessary for the core assessment system**. Conventional search, filtering, and structured data retrieval can handle evidence-to-practice mapping. RAG may provide marginal value in drafting recommendation text from gap analysis results, but these are **convenience features**, not **core functionality**. The thesis should be positioned as an assessment system that *may* include AI-assisted features, not an AI system that happens to use COBIT.

1.5 The Defensible Contribution

The student's defensible contribution is:

1. **A working web-based system that implements the official COBIT 2019 CPM rating scale** (F/L/P/N thresholds) with full aggregation from activity to practice to objective to domain
2. **Transparent scoring methodology** — since ISACA provides no PAM, the system makes its scoring assumptions explicit and configurable
3. **Evidence-to-rating traceability** — every capability rating can be traced back to specific evidence documents
4. **Gap analysis with target vs. current comparison** — visual identification of capability gaps
5. **Assessment reporting** — structured output suitable for audit documentation

This is a **system development contribution** (rancang bangun), not an AI methodology contribution. It fits squarely in the SAGE domain.

1.6 Feasibility Assessment

For an S1 Information Systems thesis, this scope is **realistic and achievable**:

- The prototype already exists and works
- The COBIT 2019 framework is publicly documented
- No backend is required (client-side SPA)
- No real organizational data is needed (mock data suffices for development)
- Evaluation can be done via expert review and usability testing

The risk is **scope creep** — trying to implement all 40 objectives, all 1,202 activities, all 11 design factors, and full AI integration. The thesis should focus on a **representative subset** (e.g., 2-3 domains, 10-15 objectives) with the architecture designed for extensibility.

1.7 The Competitive Landscape

Category	Tools	COBIT 2019 Assessment?
Enterprise GRC	ServiceNow, RSA Archer, MetricStream, Diligent, Hyperproof, Optro	No — content mapping only
ISACA's own tools	CMMI Cybermaturity Platform	No — COBIT 5, cyber-scoped
Dedicated COBIT tools	ITS Governance, Compliance	Claimed but unverified scoring

Academic prototypes	Mandiangan 2023, Anwar and Harits 2025, Noor 2021	Yes but isolated, undocumented
Open source	ControlWeave	No — NIST/ISO assessment model

The student's system would be among the first to **transparently implement and document** the COBIT 2019 CPM methodology in a web-based tool.

1.8 Methodology Recommendation

Design Science Research Methodology (DSRM) is the most appropriate methodology for this thesis. It fits the "rancang bangun" (system development) paradigm common in Indonesian S1 theses, provides clear evaluation criteria, and allows the student to claim both a **system artifact** and a **design principle** as contributions.

1.9 Thesis Positioning

The thesis should be positioned as:

"Rancang Bangun Sistem Pendukung Assessment Tata Kelola TI Berbasis COBIT 2019 dengan Pendekatan Performance Management"

With RAG/AI mentioned only as a supporting feature in specific chapters, not in the title or primary research questions.

1.10 Final Recommendation

Build the assessment system. Make the scoring methodology transparent. Use RAG only if time permits and it adds clear value. Document everything. Evaluate with experts.

The contribution is the system and its methodology, not the AI.

2. Existing Tools Comparison Table

2.1 Commercial GRC Platforms

System	Type	COBIT Version	Evidence Mgmt	Assessment Workflow	Capability Calc	Gap Analysis	Recommendations	AI/GenAI Collaboration
Diligent HighBond	Commercial SaaS	2019 (content library)	Yes	Compliance mapping only	Not confirmed	Yes	Not confirmed	Not confirmed Yes
Optro (ex-AuditBoard)	Commercial SaaS	Not stated (30+ frameworks)	Yes	Audit workflow	Not confirmed	Yes	Not confirmed	Yes (agentic) Yes
MetricStream	Commercial enterprise	Version not stated	Yes	Compliance workflow	Not confirmed	Yes (auto gap)	Yes (AI remediation)	Yes (AI-First) Yes

Hyperpro	Comme 2019 SaaS (confirm)	Yes	Control testing (Satisfied/Ga	Not confirmed	Yes	Not confirmed	Yes (Hyperpro AI)	Yes
ServiceNo GRC	Comme DIY (not platform shipped)	Yes	Custom build	Not confirmed	Custom	Custom	Yes (Now Platform)	Yes
RSA Archer	Comme None enterpri found	Yes	Custom	Not confirmed	Custom	Custom	Yes (Archer Evolv)	Yes
SAP GRC	Comme None enterpri confirm	Yes	SoD, access control	Not confirmed	Custom	Custom	Not confirmed	Yes
Thomson Reuters ONESOUR	Comme None suite found	Yes	Compliance workflow	Not confirmed	Custom	Custom	Not confirmed	Yes
Wolters Kluwer TeamMate	Comme Not in suite library	Yes	Audit test library	Not confirmed	Yes (control mapping)	Not confirmed	Not confirmed	Yes

2.2 ISACA / CMMI Tools

System	Type	COBIT Version	Evidence Mgmt	Assessment Workflow	Capability Calc	Gap Analysis	Recommendations	AI/GenAI	Collaborat
CMMI Cybermatur Platform	Commer (ISACA)	COBIT 5 (not 2019)	Yes	Risk profile then self-assessment	Yes (practice in place/not in place)	Yes (practice gap reports)	Yes (prioritized roadmap)	Not mentione	Yes (team assignment)
COBIT 2019 Tool Kit	Free Excel	2019	No	Manual	Manual	Manual	Manual	No	No

2.3 Dedicated COBIT Assessment Tools

System	Type	COBIT Version	Evidence Mgmt	Assessment Workflow	Capability Calc	Gap Analysis	Recommendations	AI/GenAI	Collaboration
ITS Governance	Service + platform	2019	Not verified	Questionnaire then scoring	Claimed (method unverified)	Yes	Yes	Not confirmed	Not verified
Complya	Comme SaaS	2019	Yes (evidence automatic	Continuous monitoring	Not confirmed	Yes	Yes (AI-assisted)	Yes (AI-assisted reviews)	Yes

Dynaflow Commerce	(>1200 controls)	No	Review and test cycles	Not confirmed	Not confirmed	Not confirmed	No	No
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2.4 Open Source Tools

System	Type	COBIT Version	Evidence Mgmt	Assessment Workflow	Capability Calc	Gap Analysis	Recommendations	AI/GenAI	Collaboration	Feedback
ControlWorks	Open source (AGPL 3.0)	2019 (among 44 frameworks)	Yes (version history, SHA-256)	Control testing (Satisfied/OT)	Not confirmed	Yes (POA and M)	Yes (remediation playbooks)	Yes (optional BYOK)	Yes (RBAC, workpapers)	Yes (Feedback)
A-COBIT-2019-Risk-Assessment	Open source (GPL-3.0)	2019	No	Risk assessment	Not confirmed	Not confirmed	Not confirmed	No	No	Comments
evaluator-ti	Open source	COBIT/IT	No	Maturity questionnaire	Yes (maturity scoring)	Yes	Not confirmed	No	No	Lessons Learned

2.5 Academic Prototypes

Study	Year	Method	COBIT Version	Capability Calc	Evidence Mgmt	Gap Analysis	Key Limitation
Mandiangan (UPN Veteran Jakarta)	2023	Agile Scrum, web-based	2019	Yes (percentage-based)	No	Yes	Replaces Excel; undocumented scoring formula
Anwar and Harits (Jurnal Informatika Utama)	2025	RAD, PHP/Java/MySQL	2019	Yes (capability score)	No	Not confirmed	Questionnaire system; limited documentation
Noor (ITTP Purwokerto)	2021	Prototyping, black-box	2019	Yes (BAI domain)	No	Not confirmed	Single domain; no evidence management
Cobvapps (UB)	—	Web app	2019	Yes (maturity)	No	Not confirmed	Prototype; limited documentation
Pamuji et al.	2022	ISO/IEC 15504-2	2019	Yes (audit assessment)	No	Not confirmed	Notes absence of PAM for COBIT 2019

2.6 Summary: What Exists vs. What is Missing

Capability	Commercial GRC	ISACA Tools	Dedicated COBIT	Open Source	Academic	This Project
COBIT 2019 content	Yes (most)	Yes (Excel)	Yes	Yes	Yes	Yes
F/L/P/N rating scale	No	No	Unverified	No	Partial	Yes
Activity-Practice-	No	No	Unverified	No	Partial	Yes

Objective aggregation	Yes	Yes	Unverified	Yes	Partial	Yes
Evidence-to-rating traceability	Partial	No	Unverified	Partial	No	Yes
Gap analysis (current vs. target)	Partial	Yes (CMMI)	Yes	Yes	Yes	Yes
Recommendation generation	Partial	Yes (CMMI)	Yes	Yes	No	Yes
Transparent scoring methodology	No	No	No	No	No	Yes
Web-based, no backend	No	No	No	No	Partial	Yes
Open source / free	No	No	No	Yes	No	Yes

3. System Requirements (Derived from COBIT 2019)

3.1 Core COBIT Assessment Functionality (Must-Have)

ID	Requirement	COBIT 2019 Source	Prototype Module	Priority
CR-01	Assessment project setup (org, period, assessor)	Step 1-3: Awareness, design, briefing	AssessmentSetup	High
CR-02	COBIT domain selection (EDM/APO/BAI/DSS/MEA)	5 domains, 40 objectives	AssessmentSetup	High
CR-03	Goals Cascade visualization (EG-AG-GM)	13 EG, 13 AG, 40 G and M	Dashboard	Medium
CR-04	Design Factors configuration	11 design factors	AssessmentSetup	Medium
CR-05	Practice/activity listing per objective	231 practices, 1,202 activities	AssessmentWorkspace	High
CR-06	Activity rating (N/P/L/F) with thresholds	CPM: less than 15 percent / 15-50 / 50-85 / above 85	AssessmentWorkspace	High
CR-07	Capability level calculation (0-5)	CMMI-based levels	CapabilityResult	High
CR-08	Aggregation: activity-practice-objective-domain	1,202-231-40-5	CapabilityResult	High
CR-09	Target level setting per objective	Step 6: Reporting	GapAnalysis	High
CR-10	Gap analysis (current vs. target)	Step 6: Strengths/opportunities	GapAnalysis	High
CR-11	Findings documentation	Step 6: Reporting	EvidenceFindings	High
CR-12	Recommendation generation	Step 6: Opportunities	Recommendations	High
CR-13	Assessment report generation	Step 6: Reporting	Dashboard	Medium

3.2 Evidence Management Functionality

ID	Requirement	COBIT 2019 Source	Prototype Module	Priority
EV-01	Evidence upload/import	Step 4: Evidence collection	EvidenceFindings	High
EV-02	Evidence categorization (policy/record/log/interview)	Step 4: Methodologies	EvidenceFindings	High
EV-03	Evidence metadata (date, source, ...)	Step 4: Validation	EvidenceFindings	High

EV-03	Evidence type, status)	Step 4: Validation	EvidenceFindings	High
EV-04	Evidence-to-practice mapping	Step 4-5: Evidence to rating	AssessmentWorkspace	High
EV-05	Evidence validation status (pending/reviewed/attached)	Step 4: Validation	EvidenceFindings	High
EV-06	Evidence search/filter	Step 4: Additional evidence	EvidenceFindings	Medium
EV-07	Evidence sufficiency indicator	Step 4: Sufficient evidence	CapabilityResult	Medium
EV-08	Evidence version control	Inter-context conflict risk	EvidenceFindings	Low

3.3 Supporting Functionality

ID	Requirement	COBIT 2019 Source	Prototype Module	Priority
SP-01	Dashboard with progress visualization	—	Dashboard	High
SP-02	Historical assessment comparison	—	Dashboard	Low
SP-03	User roles (assessor, reviewer, admin)	—	—	Low
SP-04	Collaboration features	—	—	Low
SP-05	Export/reporting (PDF/print)	Step 6: Report	Dashboard	Medium
SP-06	Assessment workflow stages	6-step process	AssessmentWorkspace	Medium

3.4 AI-Assisted Functionality (Optional)

ID	Requirement	Problem Solved	RAG Necessary?	Alternative	Priority
AI-01	Evidence retrieval/suggestion	Finding relevant evidence for a practice	No	Structured search by practice ID	Low
AI-02	Evidence-practice matching assistance	Mapping evidence to COBIT criteria	No	Manual mapping with tagging	Low
AI-03	Draft recommendation generation	Writing recommendation text from gaps	Maybe	Template-based generation	Medium
AI-04	Evidence summarization	Quick understanding of long documents	No	Abstract/summary field	Low
AI-05	Capability rating suggestion	Suggesting N/P/L/F based on evidence	No	Assessor judgment (mandatory)	Low

3.5 Requirements Traceability Matrix

COBIT 2019 6-Step Process	-> Prototype Modules
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Step 1: Awareness sessions	-> AssessmentSetup (project creation)
Step 2: Tailored governance design	-> AssessmentSetup (domain/design factor selection)
Step 3: Process owner briefing	-> AssessmentSetup (stakeholder assignment)
Step 4: Evidence collection	-> EvidenceFindings (evidence management)
Step 5: Activity rating	-> AssessmentWorkspace (N/P/L/F rating) CapabilityResult (level calculation)
Step 6: Reporting	-> GapAnalysis, Recommendations, Dashboard

4. RAG Opportunity Analysis

4.1 Use Case Evaluation

Use Case	Problem Solved	RAG Necessary?	Alternative	Risk	Human Validation	S1 Feasibility
Evidence retrieval	Find evidence relevant to a COBIT practice	No — structured search by practice ID, domain, or keyword suffices	SQL/filter query on evidence metadata	Low	Low	High
Evidence-practice matching	Suggest which evidence maps to which practice	No — manual mapping with tagging is more accurate and auditable	Tagging system with practice IDs	Low	Medium	High
Draft recommendation generation	Auto-write recommendation text from gap analysis	Maybe — templates may suffice, but RAG could produce more natural language	Template-based generation with fill-in fields	Medium	High	Medium
Evidence summarization	Summarize long evidence documents	No — assessors need to read full documents for audit reliability	Abstract field or manual summary	High	High	Low
Capability rating suggestion	Suggest N/P/L/F rating based on evidence	No — this is assessor judgment; AI suggestion would undermine audit reliability	Assessor decision with evidence reference	Very High	Mandatory	Low
Missing evidence detection	Identify what evidence is missing for a practice	No — checklist of required evidence types per practice	Structured checklist	Low	Low	High
Cross-assessment learning	Learn from past assessments to improve current	No — historical comparison is straightforward data analysis	Dashboard comparison	Low	Low	Medium

4.2 Where RAG Could Add Value (If Included)

Only one use case justifies RAG for an S1 thesis: **Draft Recommendation Generation (AI-03)**

- **Problem:** Writing clear, actionable recommendation text from gap analysis results is time-consuming
- **RAG value:** Could retrieve similar past recommendations and adapt them to current gaps
- **Input:** Gap analysis results (current level, target level, specific gaps)
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- **Output:** Draft recommendation text with references to COBIT practices
- **Risk:** Generated text may not be specific enough; requires heavy human editing
- **Human validation:** Mandatory — all AI-generated recommendations must be reviewed and approved by assessor
- **S1 feasibility:** Medium — requires building a recommendation corpus and RAG pipeline

4.3 Where RAG Should NOT Be Used

Use Case	Why RAG Is Inappropriate
Capability rating (N/P/L/F)	This is professional judgment. AI suggestion undermines audit reliability and ISACA CPM requires assessor decision.
Evidence validation	Evidence reliability depends on source, nature, circumstances — not text similarity.
Assessment scoring	Scoring must follow deterministic rules (F/L/P/N thresholds), not probabilistic generation.
Audit opinion	Legal accountability requires human assessor.

4.4 Minimum Viable RAG Integration (If Pursued)

If the student decides to include RAG as a supporting feature:

1. **Scope:** Recommendation drafting only
2. **Corpus:** COBIT 2019 framework documents + sample recommendation templates
3. **Retrieval:** BM25 + vector search (hybrid)
4. **Generation:** LLM with strict grounding in retrieved COBIT practices
5. **Evaluation:** Faithfulness score + human review by assessor
6. **Fallback:** Template-based generation if RAG confidence is low

4.5 Recommendation on RAG

For an S1 SAGE thesis, RAG is optional and should not be the primary focus. The core contribution is the assessment system. If included, RAG should be:

- Clearly scoped to one use case (recommendation drafting)
- Positioned as a "supporting feature" or "enhancement"
- Evaluated separately from the core system
- Developed only after the core system is complete

If the student cannot build a working RAG system, the thesis is still valid without it.

5. Research Methodology Comparison

5.1 Methodology Options

Method	Fits "Rancang Bangun"?	Evaluation Approach	Contribution Claim	S1 Fit
Design Science Research (DSRM)	Excellent — designed for system development	Expert review, usability testing, artifact evaluation	System artifact + design principles	Best fit
Prototyping	Good — iterative	Black-box testing, user	Working system	Good fit

Prototyping	build-test-refine	feedback	working system	Good fit
SDLC (Waterfall)	Partial — too rigid for research	System testing, requirements validation	System implementation	Weak fit
Agile Scrum	Partial — development method, not research	Sprint reviews, user stories	System implementation	Weak fit
Case Study	Partial — observation, not building	Interviews, observation	Understanding, not system	Poor fit
Action Research	Partial — participatory	Cyclical intervention	Change, not system	Poor fit

5.2 Recommended: Design Science Research Methodology (DSRM)

DSRM (Peffer et al., 2007) is the most appropriate because:

1. **It is designed for "rancang bangun"** — the primary goal is to build an artifact (the assessment system)
2. **It has clear evaluation criteria** — the artifact must be evaluated against requirements
3. **It allows dual contribution** — both the system and the design principles
4. **It is widely accepted in IS research** — especially for S1/S2 theses in Indonesian universities
5. **It accommodates iterative development** — build, evaluate, refine cycles

5.3 DSRM Process for This Thesis

Phase 1: Problem Identification

- > Literature review (done)
- > Problem: No transparent, web-based COBIT 2019 assessment tool
- > Objective: Build a system that implements CPM methodology

Phase 2: Solution Design

- > Requirements derivation (Section 3)
- > Architecture design (React 19 + TypeScript SPA)
- > Scoring methodology design (F/L/P/N + aggregation)

Phase 3: Development

- > Build prototype (already exists, needs refinement)
- > Implement scoring engine
- > Implement evidence management
- > Implement gap analysis and recommendations

Phase 4: Evaluation

- > Expert review (COBIT practitioners)
- > Usability testing (SUS questionnaire)
- > Black-box testing (functional requirements)
- > Comparison with manual assessment

Phase 5: Communication

- > Thesis write-up
- > Documentation

5.4 Evaluation Methods

Evaluation Method	What It Measures	How to Apply	S1 Feasibility
Expert review	Correctness of COBIT 2019 implementation	2-3 COBIT practitioners review scoring and methodology	High
Usability testing (SUS)	Ease of use	5-8 assessors use system, fill SUS questionnaire	High

Black-box testing	Functional correctness	Test all requirements (CR-01 to CR-13)	High
Comparison with manual	Scoring accuracy	Compare system output with manual Excel assessment	Medium
Inter-rater reliability	Consistency across assessors	Multiple assessors rate same practices, compare	Medium

5.5 What Can Be Claimed as Contribution

Contribution Type	Claim	Evidence Required
System artifact	A working web-based COBIT 2019 assessment system	Working prototype + documentation
Design principle	Transparent scoring methodology for COBIT 2019 CPM	Published methodology + expert validation
Empirical insight	Usability and usefulness for assessors	SUS scores + expert feedback
Methodological	DSRM applied to COBIT assessment	Thesis documentation

Cannot claim:

- "First COBIT 2019 assessment tool" — academic prototypes exist
- "Novel AI technique" — RAG is not the focus
- "Improves audit quality" — no controlled study
- "Replaces assessor judgment" — system is decision-support only

6. Research Gap (Conservative)

6.1 What Existing Tools Do Not Address

Gap	Evidence	Defensibility
No commercial tool implements F/L/P/N rating scale	Searched 9 commercial platforms; none document CPM rating	Strong — verified from official sources
No tool documents the 1,202-231-40-5 aggregation	Only Dynaflow productizes the 1,202 activities as a control library	Strong — verified
No transparent scoring methodology	ISACA provides no PAM; tools that claim COBIT 2019 do not document scoring	Strong — ISACA states this explicitly
Academic prototypes are isolated and undocumented	Mandiangan 2023, Anwar and Harits 2025 exist but do not document methodology	Moderate — papers exist but are hard to access
No web-based, open-source COBIT 2019 assessment tool	ControlWeave is NIST/ISO; no open-source COBIT CPM tool found	Moderate — search may have missed tools
Evidence-to-rating traceability is weak	Commercial tools have evidence management but do not link to COBIT ratings	Moderate — inferred from tool capabilities

6.2 What the Student's System Adds

Differentiator	Description	Defensibility
Transparent F/L/P/N implementation	System explicitly implements ISACA published thresholds	Strong — verifiable against ISACA documentation
Full aggregation chain	Activity-Practice-Objective-Domain with traceability	Strong — can be demonstrated
Evidence-to-rating linkage	Every rating can be traced to specific evidence	Moderate — depends on implementation quality
Open source and free	No licensing cost; accessible to students and small organizations	Strong — GitHub repository
Web-based, no backend	Client-side SPA; no server infrastructure needed	Moderate — technical choice, not unique
Documented methodology	Thesis documents scoring assumptions and limitations	Strong — thesis is the documentation

6.3 Conservative Gap Statement

"While COBIT 2019 is widely used for IT governance assessment, ISACA publishes no Process Assessment Model (PAM) for COBIT 2019, and no commercial or open-source tool transparently implements the official Fully/Largely/Partially/Not activity rating scale with full aggregation from activity to domain level. Existing academic prototypes implement partial solutions but do not document their scoring methodology or provide evidence-to-rating traceability. This project addresses this gap by developing a web-based assessment system that transparently implements the COBIT 2019 CPM methodology with full traceability."

7. Thesis Scope Options with Trade-offs

Option A: Core Assessment System Only (Recommended)

Scope: Build a web-based system that implements COBIT 2019 CPM rating, aggregation, gap analysis, and reporting. No AI.

Aspect	Evaluation
Description	Rancang Bangun Sistem Assessment COBIT 2019 dengan CPM
Complexity	Medium
Research depth	Medium
Implementation effort	Medium (prototype exists)
Evaluation effort	Medium (expert review + usability)
Data requirements	Mock data (sufficient)
S1 suitability	High — focused, achievable, defensible
Risk	Low
Contribution	System + methodology
Verdict	Best for S1 — clear scope, realistic, defensible

Option B: Assessment System + Evidence Management

Scope: Option A + full evidence management (upload, categorization, mapping, validation).

Aspect	Evaluation
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Description	Sistem Assessment + Manajemen Evidensi
Complexity	Medium-High
Research depth	Medium
Implementation effort	High
Evaluation effort	Medium-High
Data requirements	Mock evidence documents
S1 suitability	Medium-High — more work but more complete
Risk	Medium
Contribution	System + methodology + evidence framework
Verdict	Good if time permits; evidence management adds value

Option C: Assessment System + AI-Assisted Recommendations

Scope: Option A + RAG-based recommendation drafting.

Aspect	Evaluation
Description	Sistem Assessment + Rekomendasi Berbantuan AI
Complexity	High
Research depth	Medium-High
Implementation effort	Very High
Evaluation effort	High
Data requirements	COBIT docs + recommendation corpus
S1 suitability	Medium — RAG adds significant complexity
Risk	Medium-High
Contribution	System + methodology + AI feature
Verdict	Risky for S1; AI may not work well enough to be valuable

Option D: Full Assessment System + Evidence + AI

Scope: Everything — assessment, evidence management, gap analysis, recommendations, AI assistance.

Aspect	Evaluation
Description	Sistem Komprehensif Assessment + Evidensi + AI
Complexity	Very High
Research depth	High
Implementation effort	Very High
Evaluation effort	Very High
Data requirements	Full organizational data
S1 suitability	Low — too large for S1
Risk	High
Contribution	Would be impressive but unlikely to complete
Verdict	Not recommended — high risk of incomplete thesis

Option E: Assessment System + Historical Comparison

Scope: Option A + historical assessment comparison and trend analysis.

Aspect	Evaluation
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Description	Sistem Assessment + Perbandingan Historis
Complexity	Medium
Research depth	Medium
Implementation effort	Medium
Evaluation effort	Medium
Data requirements	Multiple assessment periods
S1 suitability	High — adds value without much complexity
Risk	Low
Contribution	System + methodology + trend analysis
Verdict	Good option; historical comparison is straightforward

7.1 Recommendation

Option A (Core Assessment System Only) is the recommended scope for an S1 thesis. It provides:

- Clear, bounded scope
- Defensible contribution (system + methodology)
- Realistic timeline
- Low risk
- Foundation for future work (evidence management, AI can be added later)

If the student has time and energy after completing Option A, **Option B (Evidence Management)** or **Option E (Historical Comparison)** can be added as enhancements.

8. Potential Research Question Sets

RQ Set 1: Core Assessment System (Option A — Recommended)

RQ1: *Bagaimana merancang dan membangun sistem pendukung assessment tata kelola TI berbasis web yang mengimplementasikan metodologi COBIT 2019 Performance Management secara transparan?*

RQ2: *Sejauh mana sistem yang dibangun mampu menghasilkan perhitungan capability level (0-5) yang sesuai dengan skala penilaian COBIT 2019 (F/L/P/N) berdasarkan input penilaian aktivitas?*

RQ3: *Bagaimana tingkat usability sistem pendukung assessment COBIT 2019 ini menurut perspektif assessor yang menggunakannya?*

RQ Set 2: Assessment + Evidence Management (Option B)

RQ1: *Bagaimana merancang dan membangun sistem assessment tata kelola TI berbasis web dengan manajemen evidensi terintegrasi untuk mendukung proses assessment COBIT 2019?*

RQ2: *Sejauh mana sistem yang dibangun mampu memetakan evidensi ke dalam kriteria penilaian COBIT 2019 dan menghasilkan capability level yang traceable?*

RQ3: *Bagaimana dampak penggunaan sistem ini terhadap efisiensi dan konsistensi proses assessment COBIT 2019 dibandingkan dengan metode manual?*

RQ Set 3: Assessment + AI-Assisted Recommendations (Option C)

RQ1: *Bagaimana merancang dan membangun sistem pendukung assessment tata kelola TI berbasis web dengan fitur rekomendasi berbantuan AI menggunakan pendekatan Retrieval-Augmented Generation?*

RQ2: *Sejauh mana sistem RAG yang diintegrasikan mampu menghasilkan draf rekomendasi yang grounded pada gap analysis COBIT 2019?*

RQ3: *Bagaimana dampak penggunaan fitur rekomendasi AI terhadap efisiensi penyusunan rekomendasi perbaikan tata kelola TI?*

RQ Set 4: Assessment + Historical Comparison (Option E)

RQ1: *Bagaimana merancang dan membangun sistem assessment tata kelola TI berbasis web dengan fitur perbandingan historis untuk mendukung evaluasi progres COBIT 2019?*

RQ2: *Sejauh mana visualisasi tren capability level antar-periode assessment dapat membantu pemangku kepentingan memahami progres tata kelola TI?*

RQ3: *Bagaimana tingkat usability sistem ini menurut perspektif assessor dan manajemen yang menggunakannya untuk evaluasi progres?*

RQ Set 5: Comprehensive System (Option D — Not Recommended for S1)

RQ1: *Bagaimana merancang dan membangun sistem komprehensif assessment tata kelola TI berbasis web yang mengintegrasikan assessment COBIT 2019, manajemen evidensi, dan fitur rekomendasi berbantuan AI?*

RQ2: *Sejauh mana sistem komprehensif ini mampu mendukung seluruh proses assessment COBIT 2019 dari pengumpulan evidensi hingga pelaporan?*

RQ3: *Bagaimana dampak penggunaan sistem komprehensif ini terhadap efisiensi, kualitas, dan konsistensi proses assessment tata kelola TI?*

9. Potential Thesis Titles

Primary Titles (COBIT 2019 Assessment Focus)

1. **Rancang Bangun Sistem Pendukung Assessment Tata Kelola TI Berbasis COBIT 2019 dengan Pendekatan Performance Management**
2. **Rancang Bangun Aplikasi Web Assessment Kapabilitas TI Berbasis COBIT 2019 dengan Skala Penilaian F/L/P/N**
3. **Rancang Bangun Sistem Informasi Assessment COBIT 2019 Berbasis Web dengan Kemampuan Perhitungan Capability Level**
4. **Rancang Bangun Sistem Pendukung Penilaian Tata Kelola TI Berbasis COBIT 2019 dengan Metode Design Science Research**

5. **Rancang Bangun Aplikasi Assessment COBIT 2019 Berbasis Web untuk Mengukur Capability Level Tata Kelola TI**

Titles with Evidence Management Focus

6. **Rancang Bangun Sistem Assessment COBIT 2019 Berbasis Web dengan Manajemen Evidensi Terintegrasi**
7. **Rancang Bangun Sistem Pendukung Assessment Tata Kelola TI Berbasis COBIT 2019 dengan Traceability Evidensi**
8. **Rancang Bangun Aplikasi Web Assessment COBIT 2019 dengan Kemampuan Pemetaan Evidensi ke Kriteria Penilaian**

Titles with AI/RAG as Supporting Feature

9. **Rancang Bangun Sistem Assessment COBIT 2019 Berbasis Web dengan Fitur Rekomendasi Berbantuan AI**
10. **Rancang Bangun Sistem Pendukung Assessment Tata Kelola TI Berbasis COBIT 2019 dengan Rekomendasi Menggunakan Retrieval-Augmented Generation**
11. **Rancang Bangun Aplikasi Assessment COBIT 2019 Berbasis Web dengan Dukungan Rekomendasi Berbasis RAG**

Titles with Historical/Trend Focus

12. **Rancang Bangun Sistem Assessment COBIT 2019 Berbasis Web dengan Analisis Tren Antar-Periode**
13. **Rancang Bangun Sistem Pendukung Evaluasi Progres Tata Kelola TI Berbasis COBIT 2019 dengan Perbandingan Historis**

Titles with Specific Domain Focus

14. **Rancang Bangun Sistem Assessment COBIT 2019 Berbasis Web (Studi Kasus: Domain APO pada Organisasi X)**
15. **Rancang Bangun Aplikasi Assessment Kapabilitas COBIT 2019 Berbasis Web (Studi Kasus: Domain DSS dan MEA)**

9.1 Title Recommendation

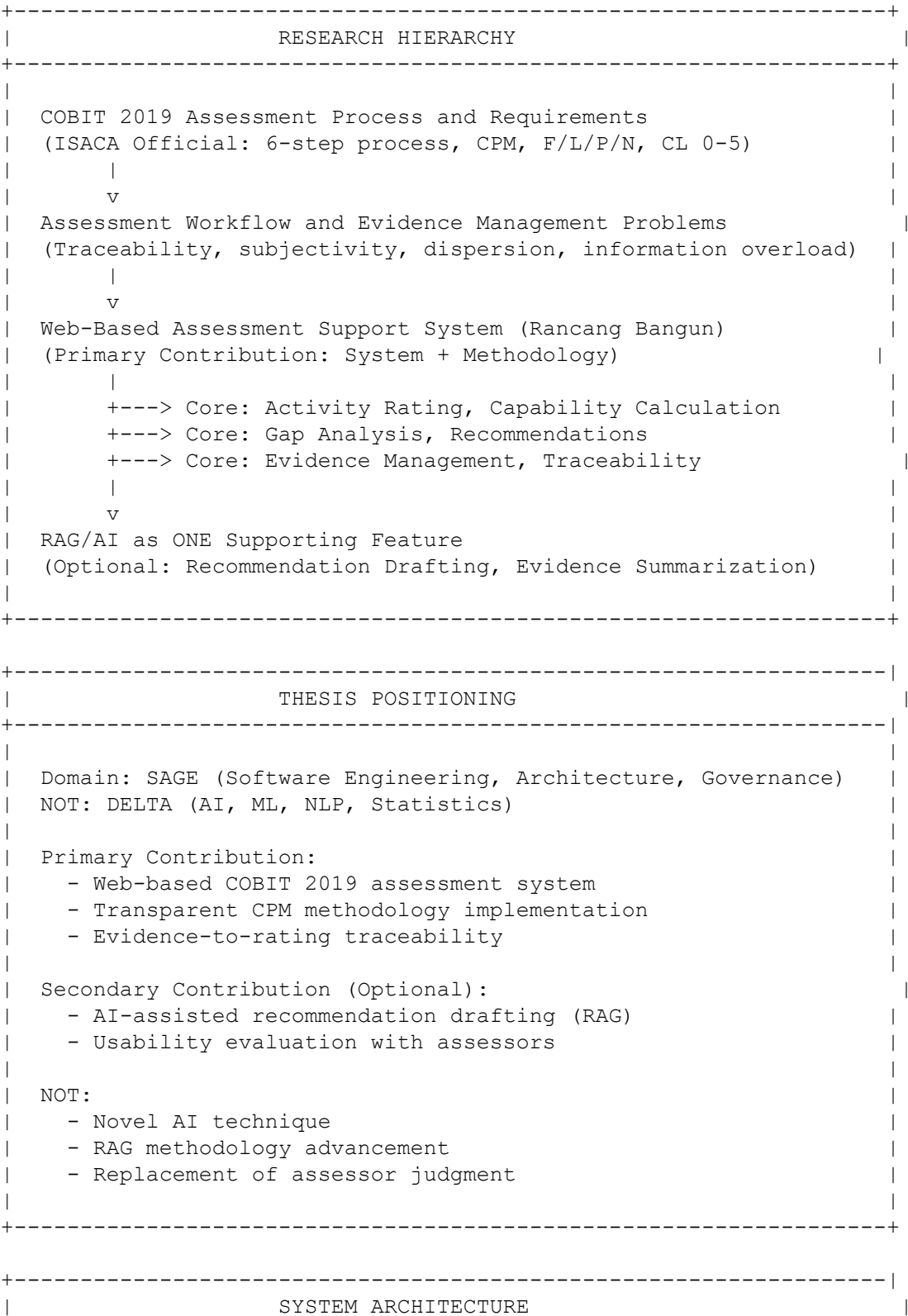
Recommended title (Option A scope):

Rancang Bangun Sistem Pendukung Assessment Tata Kelola TI Berbasis COBIT 2019 dengan Pendekatan Performance Management

This title:

- Clearly states "Rancang Bangun" (system development)
- Names COBIT 2019 as the framework
- Mentions "Performance Management" (the CPM methodology)
- Does not over-emphasize AI/RAG
- Is specific enough for S1 scope
- Can be adapted with study case or domain focus

10. Conceptual Positioning Diagram



+-----+		+-----+		+-----+	
	AssessmentSetup		AssessmentWorkspace		CapabilityResult
	- Project config		- Activity rating		- CL 0-5 calc
	- Domain select		- Evidence link		- Aggregation
	- Design factors		- N/P/L/F input		- Traceability
+-----+		+-----+		+-----+	
v		v		v	
+-----+		+-----+		+-----+	
	EvidenceFindings		GapAnalysis		Recommendations
1	- Upload/import	1	- Current vs.	1	- Gap-based
1	- Categorize	1	Target	1	generation
1	- Map to practice	1	- Visualization	1	- AI-assisted
1	- Validate	1	- Findings	1	(optional)
+-----+		+-----+		+-----+	
v		v		v	
+-----+					
Dashboard					
- Progress visualization		- Historical comparison			
- Assessment report		- Export/print			
+-----+					
+-----+					

Appendix: Key Sources

ISACA Official

- COBIT 2019 Framework: Introduction and Methodology
- COBIT 2019 Framework: Governance and Management Objectives
- COBIT 2019 Design Guide
- COBIT 2019 Implementation Guide
- ISACA Journal: "Building a Maturity Model for COBIT 2019 Based on CMMI" (2021)
- ISACA Now Blog: "Getting Creative with Maturity Models and COBIT 2019" (2022)

Commercial Tools

- Diligent HighBond: COBIT 2019 Control Libraries
- Hyperproof: Supported Frameworks (COBIT 2019 confirmed)
- MetricStream: Connected GRC
- ServiceNow GRC Community: COBIT 2019 mapping discussions
- Optro (ex-AuditBoard): Frameworks page
- ControlWeave: GitHub repository (AGPL-3.0)

Academic

- Mandiangan (2023): Rancang Bangun Sistem Informasi Capability Assessment Tools COBIT 2019
- Anwar and Harits (2025): Perancangan Sistem Kuisiener Penilaian Kapabilitas Framework COBIT 2019
- Noor (2021): Implementasi Sistem Penilaian Kapabilitas Tata Kelola TI Berbasis COBIT 2019
- Pamuji et al. (2022): Using COBIT-19 with ISO/IEC 15504-2
- Mambu et al. (2025): EDM + APO domains assessment

